👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 4 · STANDARD 6.AT.4

Understand Percent

Lesson 4-1 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can explain a percent as a rate per 100, model it on a decimal grid or tape diagram, and interpret percents greater than 100%.

Language Objective: I can explain what a percent means using the words per hundred, part, whole, and equivalent ratio.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Percent · **Per hundred** · **Decimal grid** · **Part-to-whole** · **Greater than 100%** · **Equivalent ratio** · **Tape diagram**

1. The word that means “per hundred” is _____.

2. The meaning hidden inside the word percent is _____ _____.

3. A 10-by-10 grid of 100 squares used to show a percent is a _____ _____.

4. A comparison of one part against the whole amount is a _____ comparison.

5. A percent above 100 means the amount is _____ than one whole.

6. A ratio made by multiplying both terms by the same number is an _____.

7. A bar split into equal parts that shows a percent is a _____.

Write About the Math

What percent of the gray kitten's weight does the orange kitten weigh, and how does the tape diagram show it?
¿Qué porcentaje del peso del gatito gris pesa el gatito naranja, y cómo lo muestra el diagrama de cinta?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Percent** — *Porcentaje*
- ☐ **Per hundred** — *Por cada cien*
- ☐ **Decimal grid** — *Cuadrícula decimal*
- ☐ **Part-to-whole** — *Parte a todo*
- ☐ **Greater than 100%** — *Mayor que 100 %*

Model: *150% is more than one whole, and 0.5% is a tiny piece smaller than 1%.* — Students explain 150% is more than one whole (the player beat the goal) and 0.5% is far less than one whole (a tiny boost), connecting to the idea of 100% as the whole.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The gray kitten is the ___, so its weight is ___%. The orange kitten needs one whole tape plus ___ of another. Adding the parts, ___% + 50% = ___%. A percent greater than 100% means the orange kitten weighs ___ than one whole gray kitten.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Percent
2. **Notes 2:** Per hundred
3. **Notes 3:** Decimal grid
4. **Notes 4:** Part-to-whole
5. **Notes 5:** Greater than 100%
6. **Notes 6:** Equivalent ratio
7. **Notes 7:** Tape diagram

8. **Watch for this mistake:** *A common mistake with percent is scaling only one term of the ratio: reading 7 out of 10 as 7%. A percent compares to 100, so BOTH terms must be multiplied by the same factor — 10 × 10 = 100, so 7 × 10 = 70, giving 70%. Reading it as 7% describes seats that are nearly empty when they are nearly full. A second mistake is treating a percent as an amount: 150% of a 200-point goal is 1.5 × 200 = 300 points, not 150 points.*
9. **Try It 1:** D. Twice the goal — *100% is the whole goal, so 200% is two of those — twice the goal.*
10. **Try It 2:** — *Dividing a percent by 100 tells you how many wholes it is.*